KARTIKEYA VEMULA

Analytics Engineer | dbt | BigQuery | GCP Certified | Power BI (PL-300)
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Open to relocate across India

PROFILE

GCP-certified Analytics Engineer with hands-on **SQL, Python, dbt, and BigQuery** experience building **warehouse-backed KPI marts**, tested transformation layers, and decision-support reporting pipelines. Skilled at translating raw datasets into **business-ready analytics, root-cause insights, and stakeholder recommendations** across customer intelligence, product analytics, and risk analytics.

CERTIFICATIONS

- Google Cloud Certified: Professional Data Engineer
- Microsoft PL-300: Power BI Data Analyst Associate
- dbt Fundamentals
- Cisco Data Analytics Essentials
- Google Cloud Certified: Associate Cloud Engineer
- Snowflake Data Warehousing Workshop
- Google Analytics Certification
- Kaggle Advanced SQL

CORE SKILLS

Analytics Engineering & Cloud: dbt Core, **BigQuery**, **GCP (Professional Data Engineer)**, Snowflake, PostgreSQL, Docker, ETL/ELT Pipelines, Star Schema, Dimensional Modeling, Data Quality Testing

SQL & Python: SQL, **Python (pandas, NumPy, SciPy, scikit-learn, XGBoost)**, Data Cleaning, Data Validation, Root Cause Analysis, Anomaly Detection

BI & Visualization: Power BI (PL-300), DAX, Power Query, Plotly Dash, Streamlit, Looker Studio, Dashboard Development, KPI Reporting, Executive Reporting

Professional: Stakeholder Communication, Written Documentation, Cross-Functional Collaboration, Statistical Analysis, A/B Testing, Product Analytics

EXPERIENCE

Graduate Research Assistant, Risk Analytics — UMBC, Baltimore — **Oct 2025 – Jan 2026**

Project: Extreme Value AQI Risk Modeling, Baltimore

- Built an **EPA AirData ETL pipeline** converting raw air-quality data into a validated **6,573-record dataset (2006–2023)** across AQI, pollutant, and meteorological variables.
- Designed structured **data quality checks** across missing values, outliers, and station-level reporting inconsistencies to keep the dataset **defensible for risk modeling**.
- Modeled pollutant risk using **XGBoost (RMSE 8.99, R^2 0.852)**, providing a reliable baseline for expected AQI behavior.
- Applied **Extreme Value Theory (Generalized Pareto Distribution)** across **98 AQI exceedance events**, quantifying a **6.65% conditional tail-risk probability** for extreme pollution episodes.
- Translated model outputs into **stakeholder-ready risk reporting**, framing exceedance likelihood and tail-risk for non-technical monitoring and resource prioritization.

PROJECTS

Customer Intelligence Data Warehouse

Python, PostgreSQL, Docker, dbt Core, SQL, Star Schema, Power BI, DAX, Power Query

- Converted noisy retail transactions into **tested dbt marts** for executive KPIs, CLV, repeat purchase, and country revenue — governing **£3,279,364.61 in revenue** across **29,172 invoices** and **5,047 customers** with **14 models and 41 dbt tests**.
- Modeled repeat-purchase and customer-value intelligence: **3,394 repeat buyers (67.25% repeat rate)**, a **top customer worth £62,131.42**, and **top-10 customers ≈ £383,997**.
- Built country revenue marts showing **UK concentration at £2,782,932.74 (84.86% of total)**, enabling geographic performance monitoring.

- Added **product return-audit logic** separating merchandise from operational codes, preventing fees and returns from distorting product decisions.
- Delivered a **published interactive Power BI dashboard** unifying KPIs, repeat purchase, customer value, and country revenue.

KPI Reliability and Diagnostic Engine

Python, PostgreSQL, dbt Core, SQL, Streamlit, Anomaly Detection, Root Cause Analysis

- Built a **KPI diagnostic engine** monitoring **121,500 metric rows** across **5 KPIs and 5 dimensions over 150 days**, generating **12 incident records** with severity, confidence, and root-cause segment.
- Diagnosed a **CRITICAL latency incident (158.9 ms → 343.3 ms)**, with RCA isolating **North America / Desktop / Chrome as 100% of excess latency**.
- Surfaced a second incident where **APAC / Mobile / Safari drove a 44.6% conversion-rate shortfall**, directing triage to the highest-impact segment.
- Benchmarked threshold, rolling Z-score, and residual detectors — selecting **residual detection** after improving **F1 to 0.5485** and cutting **false positives from 2,958 to 563**.

Instacart Intelligence Platform

Python, PostgreSQL, Docker, dbt Core, SQL, Power BI, Cohort Analysis, Basket Analysis

- Engineered PostgreSQL and dbt marts over **3,421,083 orders** and **33,819,106 order-item rows** (**206,209 customers, 49,688 products**) to model lifecycle, retention, and basket affinity.
- Built behavioral segmentation: **69,635 Moderate Engagement**, **62,818 Loyal Routine**, and **44,873 At Risk** customers for lifecycle targeting.
- Modeled order-sequence retention showing continuation **dropped to 53.70% by order 10**, giving a concrete milestone for reorder nudges.
- Analyzed product performance — **Produce the top department** and **Banana the top SKU at 491,291 orders (84.51% reorder)** — informing merchandising.
- Packaged findings into a **published Power BI dashboard** and memo, with **basket affinity analysis** separating broad cross-sell pairs from niche high-lift recommendations.

Growth Funnel Intelligence Engine

Python, SQL, PostgreSQL, Plotly Dash, Funnel Analysis, Product Analytics, A/B Testing

- Modeled **100K GA4 ecommerce events** across **34,984 sessions** into session-level funnel stages, identifying a **75.41% Session Start to Product View drop-off** as the highest-leverage area.
- Built SQL-based funnel, attribution, and cohort views across channel, device, and engagement depth, separating **conversion quality from raw volume**.
- Found **Direct as the highest-converting channel at 3.40%** and a **High-Intent (20+ min) cohort at 25.21%**.
- Evaluated A/B results with two-proportion z-tests — **Control 1.75% vs Variant 1.65%, −5.49% lift, p=0.48** — and **recommended no rollout**.

EDUCATION

M.S. Data Science — University of Maryland Baltimore County (UMBC), USA | QS Ranked, R1 Research University — **2024 – 2026**

GPA: 3.78 / 4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science — AI & ML Specialization — GITAM University, India | 8.89 GPA, First Class with Distinction — **2020 – 2024**

EXTRA-CURRICULAR & LEADERSHIP

- **Mentored peers** across Mathematics, Digital Logic, Data Communication, Design Thinking, and Introduction to Data Analysis & Machine Learning.
- Led a **3-part UMBC peer-teaching series** on machine learning — terminology, model training, technical explanation, and real-world case studies.
- Supported **chess club event coordination** and drafted social media content for the student organizing team.